📘 Enterprise Machine Learning MCQs (Detailed, No Answers)

Model Deployment & Infrastructure

1. In enterprise ML, which deployment strategy minimizes downtime during model updates?
- A. Blue-Green Deployment
- B. Shadow Deployment
- C. Canary Deployment
- D. Rolling Restart
2. Which enterprise tool is most commonly used for orchestrating ML workflows across distributed systems?
- A. Apache Airflow
- B. Jenkins
- C. GitHub Actions
- D. Terraform
3. Which approach ensures **version control** of ML models in production?
- A. Model Registry
- B. Feature Store
- C. Data Lake
- D. CI/CD Pipeline

Feature Engineering & Data Handling

4. Which technique is best suited for handling **imbalanced datasets** in enterprise fraud detection systems?
- A. Random Oversampling
- B. SMOTE (Synthetic Minority Oversampling Technique)
- C. Class Weight Adjustment
- D. All of the above
5. In enterprise ML, which method is most effective for reducing **dimensionality** in high-volume datasets?
- A. PCA (Principal Component Analysis)
- B. One-Hot Encoding
- C. Min-Max Scaling

- D. Logistic Regression

6. Which enterprise practice ensures **data lineage tracking** for ML pipelines?

- A. Metadata Management
- B. Hyperparameter Tuning
- C. Model Ensembling
- D. Batch Normalization

Model Monitoring & Governance

7. Which metric is most relevant for detecting **data drift** in production ML systems?

- A. KL Divergence
- B. RMSE
- C. F1 Score
- D. ROC-AUC

8. In enterprise ML governance, which framework is often used to ensure **ethical AI practices**?

- A. ISO 27001
- B. NIST AI Risk Management Framework
- C. COBIT
- D. ITIL

9. Which monitoring approach is best for detecting **real-time anomalies** in ML predictions?

- A. Batch Monitoring
- B. Streaming Analytics
- C. Offline Evaluation
- D. Cross-Validation

Scalability & Performance

10. Which distributed ML training technique reduces **communication overhead** across nodes?

- A. Data Parallelism
- B. Model Parallelism
- C. Federated Learning
- D. Gradient Compression

11. Which enterprise ML framework is optimized for **large-scale GPU training**?

- A. Scikit-Learn
- B. TensorFlow
- C. PyTorch Distributed
- D. XGBoost

12. In enterprise ML, which caching strategy improves **real-time inference speed**?

- A. Redis In-Memory Cache
- B. SQL Indexing
- C. Hadoop HDFS
- D. Kafka Queues

Bias, Fairness & Compliance

13. Which fairness metric evaluates whether different demographic groups receive similar positive outcomes?
- A. Equal Opportunity
- B. Demographic Parity
- C. Predictive Parity
- D. Calibration

14. Which enterprise practice ensures compliance with **GDPR's "Right to Explanation"** in ML models?
- A. SHAP (SHapley Additive Explanations)
- B. LIME (Local Interpretable Model-agnostic Explanations)
- C. Counterfactual Explanations
- D. All of the above

15. Which bias mitigation technique is most effective during **pre-processing**?
- A. Re-weighting samples
- B. Adversarial debiasing
- C. Fairness constraints in training
- D. Post-hoc calibration

Cloud ML & Enterprise Services

16. Which cloud service provides **end-to-end ML lifecycle management** including training, deployment, and monitoring?
- A. AWS SageMaker
- B. Azure ML Studio
- C. Google Vertex AI
- D. All of the above

17. Which enterprise ML service supports **federated learning** across distributed devices?
- A. TensorFlow Federated
- B. PyTorch Lightning
- C. MLflow
- D. Kubeflow

18. Which cloud ML practice ensures **cost optimization** during large-scale training?
- A. Spot Instances

- B. Auto-Scaling Clusters
- C. Hybrid Cloud Deployment
- D. All of the above

Security & Risk Management

19. Which enterprise ML security risk involves attackers manipulating training data to degrade model performance?

- A. Model Inversion Attack
- B. Data Poisoning Attack
- C. Adversarial Example Attack
- D. Membership Inference Attack

20. Which enterprise ML security measure prevents **unauthorized model access**?

- A. Role-Based Access Control (RBAC)
- B. Encryption at Rest
- C. API Gateway Authentication
- D. All of the above